



# University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

## Programming Fundamental ~ Lab Project

### Project Information

|                 |                                   |
|-----------------|-----------------------------------|
| Project Name    | Quiz Generator System             |
| Course Title    | Programming Fundamental - PF LAB  |
| Submission Date | <b>12-01-2026</b>                 |
| Faculty         | Faculty Of Information Technology |



# University of Central Punjab

*(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)*

**FACULTY OF INFORMATION TECHNOLOGY**



# University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

## Program Description

The Quiz Generator System is a menu-driven C++ console application that allows users to attempt quizzes from multiple subjects at different difficulty levels. The system reads questions from external text files, randomizes them for each attempt, calculates results in real time, assigns grades, and stores results permanently in a file.

This project demonstrates strong use of Programming Fundamentals concepts such as file handling, structures, arrays, loops, conditional statements, input validation, and modular programming. It also enhances user experience through colored output, sound effects, and timed questions.

## Key Feature:

### 1. Multi-Subject Quiz System

- Supports multiple subjects:
  - Programming Fundamentals (PF)
  - Digital Logic and Design (DLD)
  - Mathematics (Maths)
- Each subject has separate question files
- Easy to add new subjects by adding files

### 2. Difficulty Levels

- Easy
- Medium
- Hard
- Difficulty level affects question selection and visual feedback

### 3. File-Based Question Management

- Questions are stored in external .txt files
- File paths are selected dynamically based on subject and difficulty
- Questions can be updated without changing the source code

### 4. Structured Question Handling

- Uses a `struct Question` containing:
  - Question statement
  - Four multiple-choice options
  - Correct answer
- Ensures organized and readable question management

### 5. Randomized Quiz Questions

- Questions are shuffled using randomization
- Each quiz attempt presents questions in a different order



# University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

**FACULTY OF INFORMATION TECHNOLOGY**

- Maximum of 10 questions per quiz attempt

## 6. Score, Grade, and Evaluation System

- Correct answers are counted
- Final grade is calculated based on percentage
- Grades range from A+ to F
- Grade-based feedback is shown with colors and messages

## 7. Result Storage and Viewing

- Results are saved in `results.txt`
- Stored information includes:
  - Player Name
  - Player ID
  - Subject
  - Difficulty
  - Total Questions
  - Attempted Questions
  - Correct Answers
  - Grade
- Users can view previous results from the main menu

## 8. User-Friendly Menu Interface

- Main menu options:
  - Start Quiz
  - Instructions
  - View Results
  - Exit
- Clean screen transitions using `system("cls")`
- Simple and easy navigation

## 9. Input Validation

- Player name validation
- Subject and difficulty selection validation
- Answer validation (only A–D allowed)
- Timer-based answer input handling

## 10. Enhanced User Interaction

- Colored text output using Windows API
- Sound effects using `Beep()`
- Timed questions using system clock
- Delays using `Sleep()` for better readability



## Core Programming Concepts Used

### 1. Structures

- Used to group question data (statement, options, correct answer)

### 2. Arrays

- Used to store multiple quiz questions dynamically

### 3. Loops

- `for` loop for question display
- `while` loop for file reading
- `do-while` loop for menu control

### 4. Conditional Statements

- `if-else` statements for validation and grading
- `switch-case` for menu handling

### 5. Functions (Modular Programming)

- `startQuiz()` – controls quiz execution
- `instructions()` – displays rules and grading system
- `viewResults()` – displays saved quiz results
- `getFileName()` – selects appropriate question file
- `calculateGrade()` – calculates grade
- `gradeComment()` – displays grade feedback
- `setColor()` – changes console text color

### 6. File Handling

- `ifstream` used to read quiz questions from files
- `ofstream` used to store quiz results permanently
- Ensures data persistence

## System and Library Usage

### 1. Standard Libraries

- `<iostream>` – input/output
- `<fstream>` – file handling



# University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

- `<string>` – string operations
- `<cctype>` – character validation
- `<cstdlib>` and `<ctime>` – randomization and time

## 2. Windows-Specific Functions

- `system("cls")` – clear console
- `Sleep()` – execution delay
- `Beep()` – sound feedback
- `SetConsoleTextAttribute()` – colored output
- `_kbhit()` and `_getch()` – timed input handling

## Program Flow (Core Mechanics)

1. Program starts and displays main menu
2. User selects an option
3. Player enters name and gets a unique ID
4. Subject and difficulty are selected
5. Questions are loaded from file
6. Questions are shuffled randomly
7. Quiz starts with timed questions
8. User answers MCQs
9. System evaluates answers
10. Final result and grade are displayed
11. Result is saved to file

## Conclusion

The Quiz Generator System is a complete console-based application that effectively applies Programming Fundamentals concepts. It provides an interactive learning experience, ensures data persistence through files, and maintains clean program structure using functions and structures. This project is suitable for academic evaluation and demonstrates practical understanding of C++ programming.